

Tunable B-mode Despeckling via Image Saliency and Adaptive Stochastic Resampling

Background, Motivation and Objective

The presence of speckles is characteristic of ultrasound images. These speckles, which are attributed to pixelwise variations in the coherence of the pulse-echo summation process, may obscure image interpretation and diagnostics. Despeckling filters are needed to resolve this problem. Deep learning has perhaps demonstrated efficacy at this task, but it cannot be readily generalized to out-of-distribution scenarios. In this work, we present a novel stochastic image processing framework for tunable despeckling of B-mode images, enabling sonographer control of despeckling level.

Statement of Contribution/Methods

Our framework (see Fig. A) first generates an image saliency map of the beamformed image, with identical dimension. For this task, saliency was quantified in the form of local pixel variance using a region size equal to the imaging wavelength. In the next stage, the framework randomly selects pixels from the original image with selection probability proportional (via tunable multiplier β) to saliency. It then regenerates a full image based only on those pixels using accumulated neighboring intensities. Lastly, averaging over multiple realizations of random selection was performed to increase the framework's consistency. We tested the efficacy of our proposed framework on pre-beamformed linear array (L14-5) images (5 MHz, 1 cycle, -15° : 15° compounded plane waves) acquired from an *in vitro* contrast/resolution phantom (Fig. B) and an *in vivo* human carotid/thyroid imaging scenario (Fig. C).

Results/Discussion

Fig. D-I show images for three different despeckling multipliers ($\beta = 0.1, 0.25, 0.45$). We achieved despeckling without sacrificing resolution in the phantom's point targets or the tissue boundaries *in vivo*. This preservation of resolution is quantified via the lateral (LR) and axial (AR) resolution (top image row orange text), with <0.03 mm difference compared to the original image. The imaging contrast improvement was assessed using the generalized contrast-to-noise ratio (gCNR) and contrast ratio (CR) in the central lesion *in vitro* and the carotid lumen *in vivo* (green arrows and numbers), showing gCNR approaching 1 as β increased. Overall, the tunability and resolution of our framework are both advantages in the potential translation to clinical operation, allowing sonographers to optimize image interpretability via tunable despeckling.

